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Dear Editors:

We would like to submit the paper "*Genetic Convergence Analysis of CRISPR Perturbations Deciphers Gene Functional Similarity*" by Tianyu Zhang, Ergan Shang, and Kathryn Roeder for possible publication in *Cell Press*.

Our paper introduces XConTest, a novel method for analysis of perturb-seq data to identify perturbations with convergent downstream effects.

This is a critical question in the field. Quoting Rood et al (Cell 2024): "Complex molecular circuits govern the function of cells .... Despite intense study, the systematic reconstruction of causal models of such circuits, from which we can predict their behavior and manipulate cells rationally, has remained challenging for decades." Recent breakthrough, such as Perturb-seq, open up the possibility of answering these questions. However, without suitable analytical techniques, this opportunity will be wasted.

- Pooled CRISPR screens with single-cell RNA sequencing readout (Perturb-seq) provide a key technique to determine the functionality of a gene by directly perturbing the DNA of the gene. Genetic convergence refers to the phenomenon where CRISPR disruptions of different genes lead to a similar downstream outcome. Thus understanding convergence is equivalent to understanding function.
- While several heuristic methods are available, none of them provide reliable inferential techniques. Indeed the results are often depicted with visual displays. With XConTest we provide reliable p-values, which will further rigorous scientific inquiries.
- Our framework employs a novel sparsity-inducing group Lasso formulation that explicitly penalizes weak effects. In this manner it provides a statistically rigorous measure of functional similarity. By restricting convergence to genes with stronger effects, we enhance the interpretability of our results.
- We provide an in-depth investigation of a perturbation study targeting a large number of immune-related genes. Our findings conform to known features of these genes and map them back to known functional pathways. Convergent genes reveal convergence on gene modules that are known to be risk factors for immune-related phenotypes. And an investigation of GWAS hits on the targets of the converging perturbations confirms that the discovered relationship plays out in meaningful biological terms. The analysis also generates some intriguing hypotheses that would benefit from future investigation.

We believe our work provides reliable inference methods and tools for biology and biomedicine applications in general omics.

We confirm hereby that this research is original. It has not been published, nor is it currently under consideration for publication elsewhere. The paper is not part of a consortium.

Thank you very much for considering our paper.

Sincerely,

Kathryn Roeder